Ciências da Computação

Aprendizagem Computacional

Decision trees

- Meaningful
- Easy to interpret
- Decision nodes
- Leaves correspond to a class
- Created by splitting on the values of attributes

Adequacy condition

There can be no two instances with the **same** attribute values and **different** classes

Top-Down Induction of Decision Trees

- **If** all instances belong to the same class
- **Then** return the class
- **Else**
  1. Select attribute A to split on that was not used before
  2. Separate instances by attribute values of A
  3. Return a tree where the root node is choice on A and has a subtree by applying the algorithm recursively for each value of A

Entropy

Entropy

$E = -\sum_{i=1}^{K} p_i \log_2 p_i$

Average entropy when splitting using attribute A

$E_A = \sum_{v \in A} \frac{\#\text{ instances where A = v}}{\#\text{ instances}} E_{A=v}$

Information gain

$\text{Information Gain} = \text{Initial Entropy} - E_A$

Split on information gain

- Choose attribute that **maximizes** the *information gain*
- Splitting on an attribute gives a reduction on the mean entropy
- The attribute will be used as the root of the tree
- Each subtree will correspond to each attribute value

Example

Outlook	Temperature	Humidity	Wind	Play Tennis?
Overcast	Hot	Normal	Weak	Yes
Overcast	Mild	High	Strong	Yes
Sunny	Mild	Normal	Strong	Yes
Rain	Mild	Normal	Weak	Yes
Sunny	Cool	Normal	Weak	Yes
Overcast	Cool	Normal	Strong	Yes
Rain	Cool	Normal	Weak	Yes
Rain	Mild	High	Weak	Yes
Overcast	Hot	High	Weak	Yes
Rain	Cool	Normal	Strong	No
Sunny	Hot	High	Strong	No
Sunny	Hot	High	Weak	No
Rain	Mild	High	Strong	No
Sunny	Mild	High	Weak	No

Example — Initial Entropy

Counts

- 14 examples
- 9 Yes / 5 No

Entropy

$E = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14} = 0.94$

Outlook

Overcast

- 4 examples
- 4 Yes

$E = 1 \log_2 1 = 0$

Sunny

- 5 examples
- 2 Yes / 3 No

$E = -\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} = 0.971$

Rain

- 5 examples
- 3 Yes / 2 No

$E = -\frac{3}{5} \log_2 \frac{3}{5} - \frac{2}{5} \log_2 \frac{2}{5} = 0.971$

$E_{\mathrm{Outlook}} = \frac{4}{14} \times 0 + \frac{5}{14} \times 0.971 + \frac{5}{14} \times 0.971 = 0.694$

Temperature

Cool

- 4 examples
- 3 Yes / 1 No

$E = -\frac{3}{4}\log_2\frac{3}{4} - \frac{1}{4}\log_2\frac{1}{4} = 0.811$

Mild

- 6 examples
- 4 Yes / 2 No

$E = -\frac{4}{6}\log_2\frac{4}{6} - \frac{2}{6}\log_2\frac{2}{6} = 0.918$

Hot

- 4 examples
- 2 Yes / 2 No

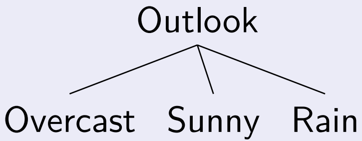
$E = -\frac{2}{4}\log_2\frac{2}{4} - \frac{2}{4}\log_2\frac{2}{4} = 1$

$$E_{\text{Temperature}} = \frac{4}{14} \times 0.811 + \frac{6}{14} \times 0.918 + \frac{4}{14} \times 1 = 0.911$$

Attribute Choice

Attribute	Mean entropy
Outlook	**0.694**
Temperature	0.911
Humidity	0.788
Wind	0.892

Tree



Outlook = Overcast

Table

Outlook	Temperature	Humidity	Wind	Play
Overcast	Hot	Normal	Weak	Yes
Overcast	Mild	High	Strong	Yes
Overcast	Cool	Normal	Strong	Yes
Overcast	Hot	High	Weak	Yes

Outlook = Sunny

Table

Outlook	Temperature	Humidity	Wind	Play
Sunny	Mild	Normal	Strong	Yes
Sunny	Cool	Normal	Weak	Yes
Sunny	Hot	High	Strong	No
Sunny	Hot	High	Weak	No
Sunny	Mild	High	Weak	No

Outlook = Sunny

Entropy

Attribute	Entropy
Outlook	0.971
Temperature	0.4
Humidity	**0**
Wind	0.951

Outlook = Rain

Table

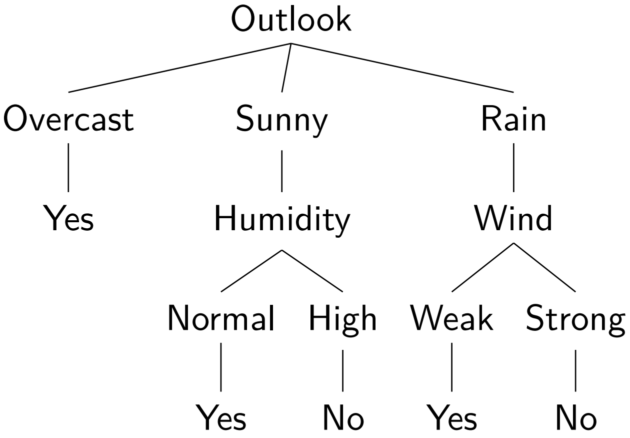
Outlook	Temperature	Humidity	Wind	Play
Rain	Mild	Normal	Weak	Yes
Rain	Cool	Normal	Weak	Yes
Rain	Mild	High	Weak	Yes
Rain	Cool	Normal	Strong	No
Rain	Mild	High	Strong	No

Outlook = Rain

Entropy

Attribute	Entropy
Outlook	0.971
Temperature	0.951
Humidity	0.951
Wind	**0**

Result


Outlook
Overcast
Sunny
Rain
Yes
Humidity
Wind
Normal
High
Weak
Strong
Yes
No
Yes
No

Transforming the Tree into Rules

```
If Outlook = Overcast Then Play = Yes
If Outlook = Sunny and Humidity = Normal  Then Play = Yes
If Outlook = Sunny and Humidity = High  Then Play = No
If Outlook = Rain and Wind = Weak  Then Play = Yes
If Outlook = Rain and Wind = Strong  Then Play = No
```

Alternative form of computing entropy

1. Subtract $f \log_2 f$ for all frequencies
2. Sum $s \log_2 s$ for all row sums
3. Divide by total number of observations

Frequencies

Values	Yes	No	Sum
Overcast	4	0	4
Sunny	2	3	5
Rain	3	2	5

Entropy

$$E_{new} = \frac{-4\log_2 4 - 2\log_2 2 - 3\log_2 3 - 3\log_2 3 - 2\log_2 2 + 4\log_2 4 + 5\log_2 5 + 5\log_2 5}{14} = 0.694$$

Gini Index of Diversity

- Measure of impurity
- A value of 0 means that all classifications are the same
- Largest value is $1 - \frac{1}{K}$ when the instances are evenly distributed between classes
- Splitting on an attribute gives a reduction on the Gini Index

Gini Index

$GI = 1 - \sum_{i=1}^{K} p_i^2$

Example

- 14 examples
- 9 of class **Yes**
- 5 of class **No**

Gini Index

$$G_{old} = 1 - (\frac{9}{14})^2 - (\frac{5}{14})^2 = 0.459$$

Computing Gini Index using frequency tables

- **For each** attribute value
  1. **Sum** the squares of the frequencies **for each** class
  2. **Divide** by the frequency **for each** attribute value
- **Sum** each subtotal and **divide** by the number of examples
- **Gini Index** is one minus that sum

Outlook

Frequencies

Values	Yes	No	Sum
Overcast	4	0	4
Sunny	2	3	5
Rain	3	2	5

Gini

$$\begin{aligned} \text{Overcast} \quad & \frac{4^2+0^2}{4} = 4 \\ \text{Sunny} \quad & \frac{2^2+3^2}{5} = 2.6 \\ \text{Rain} \quad & \frac{3^2+2^2}{5} = 2.6 \end{aligned}$$

$$G_{new} = 1 - \frac{4+2.6+2.6}{14} = 0.343$$

Attribute Choice

Attribute	Gini
Outlook	**0.343**
Temperature	0.440
Humidity	0.367
Wind	0.429

Outlook = Sunny

Attribute	Gini
Outlook	0.48
Temperature	0.2
Humidity	**0**
Wind	0.467

Outlook = Rain

Attribute	Gini
Outlook	0.48
Temperature	0.467
Humidity	0.467
Wind	**0**

Gain Ratio

Split Information Entropy corresponding to attribute value distribution

$$\text{Gain Ratio} = \frac{\text{Information Gain}}{\text{Split Information}}$$

Example for Outlook

- 4 examples of Overcast
- 5 examples of Sunny
- 5 examples of Rain

$E_{before} = 0.94$

$E_{Outlook} = 0.694$

$$\text{Gain Ratio} = \frac{0.94 - 0.694}{-\frac{4}{14}\log_2\frac{4}{14} - \frac{5}{14}\log_2\frac{5}{14} - \frac{5}{14}\log_2\frac{5}{14}} = \frac{0.94 - 0.694}{1.577} = 0.156$$

Discussion

Split Information

- It is zero if all observations belong to the same class
- For the same number of attribute values, the largest split information occurs for a uniform distribution
- For the same number of uniformly distributed instances, split information increases with more attribute values
- The largest value occurs when there are many attribute values, all equally frequent

Gain Ratio

- Information gain tends to select attributes with a large number of values
- Gain ratio substantially reduces this bias

Attribute Choice

Attribute	Gain Ratio
Outlook	**0.156**
Temperature	0.019
Humidity	0.152
Wind	0.049

Attribute Choice

Outlook = Sunny

Attribute	Gain Ratio
Outlook	0
Temperature	0.375
Humidity	**1**
Wind	0.021

Outlook = Rain

Attribute	Gain Ratio
Outlook	0
Temperature	0.021
Humidity	0.021
Wind	**1**

Dealing with clashes in the training set

- Happens when training set is not *adequate*
- Solutions include:
  - Prune branches with mixed classifications
  - Use majority voting
  - Class threshold: Use majority vote if a given percentage of the classifications belongs to the majority class, otherwise prune the branch

Tree overfitting

- A deeper tree is more specialized
- Every time an algorithm splits on an attribute, it creates a more *specialized* model
- If data contains *noise* or *irrelevant* attributes, the resulting tree will have problems generalizing
- Even if that is not the case, the tree can be:
  - Overly complex
  - Have difficulty generalizing
- In order to reduce overfitting, we must *sacrifice* classification accuracy on the training set by pruning the tree

Pruning

Pre-pruning
: prevent the generation of the full tree, by not creating *non-significant* branches

Post-pruning
: generate the full tree and then remove *non-significant* branches

Pruning

Pre-pruning

Subtree is usually substituted by leaf using *majority voting*

Size Cutoff
: prune when number of instances is less than a given value

Maximum Depth Cutoff
: prune when the branche's length reaches a given value

Independence
: class distribution is independent of the avalailable features (e.g., using χ^2 test)

No improvement
: impurity measures do not improve

Post-pruning

- Convert to rules and then prune them
- Replace some subtrees by leaf nodes
  - Reduced Error Pruning
  - Pessimistic Error Pruning

Post-pruning

Reduced Error Pruning

- Needs *unseen* examples to estimate errors
- Iteratively prune subtree that makes the most errors until further pruning increases mis-classification rate

Pessimistic Error Pruning

- Avoids using validation set
- Prune unless estimated error of subtree is more than one standard error below estimated error for pruned leaf

Pessimistic Error Pruning

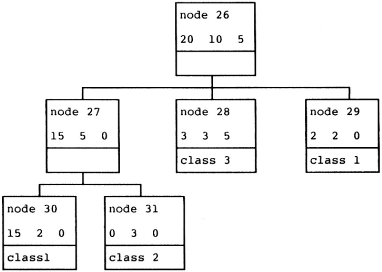
$n'(t) = e(t) + \frac{1}{2}$

$n'(T_t) = \sum e(i) + \frac{N_T}{2}$

$SE = \sqrt{\frac{n'(T_t) \times (N(t) - n'(T_t))}{N(t)}}$

Prune if if $n'(T_t) + SE > n'(t)$

Pessimistic Error Pruning



Pruning criterion

- 35 examples
- 4 leaves
- error if pruning is 15
- error of each leaf is 2, 0, 6 and 2, respectively

$n'(t) = 15 + \frac{1}{2} = 15.5$

$n'(T_t) = (2+0+6+2)+4/2 = 12.0$

$SE = \sqrt{\frac{12\times(35-12)}{35}} = 2.8$

Don't prune because

$n'(T_t) + SE = 12 + 2.8 < n'(t)$